

TAF

A Semantic Governance Framework for Assessing Article 15 Safeguards in AI-Based Digital Forensic Tool Procurement

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Demo

Three contexts, evidence gaps and human review.

One boundary throughout: evidence alignment supports expert review, not a compliance verdict.

Trustworthy AI Forensics (TAF)

MTALO: Mapping Trustworthy AI to Law Ontology. Developed using Linked Open Terms (LOT).

Why digital forensics?

Purchasing a tool can affect evidence handling, integrity and the defensibility of investigations.

The gap addressed

Fragmented legal requirements and procurement evidence make a traceable forensic review difficult.

Our approach

TAF structures review. MTALO links the concepts. The checker reports documentary matches and gaps.

Who uses the output?

Procurement, legal/compliance, cybersecurity and digital-forensics teams reviewing a proposed purchase.

THREE PROCUREMENT CONTEXTS

1. Provider → Deployer using GPAI

2. Provider → Deployer without GPAI

3. Deployer procures GPAI directly

GPAI = general-purpose AI. These are alternative scenarios, not a sequence or an automatic legal classification.

Accuracy

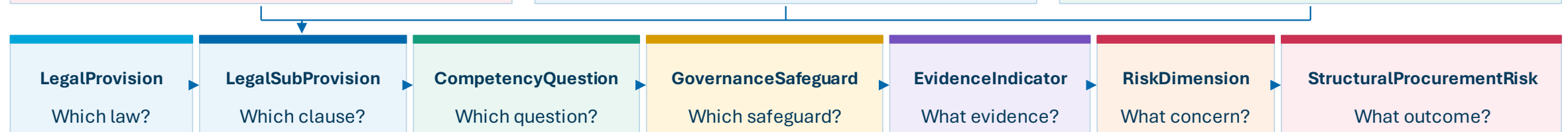
Art. 15(1), (3): declared levels and metrics

Robustness

Art. 15(1), (4): faults, resilience and feedback loops

Cybersecurity

Art. 15(1), (5): unauthorised interference and attacks



Article 15 is a high-risk-system anchor. Applicability and the final procurement decision remain human-reviewed.

Procurement claims need supporting evidence

Forensic review asks what a tool claim means for the handling and interpretation of evidence.

What a supplier may say

“accurate”
“secure”
“transparent”
“AI-powered”
“compliant”

What reviewers examine

Declared performance and limitations

Robustness and security documentation

Chain of custody, data integrity and auditability

How the output is used

Locate the supporting passage.

Ask the supplier for missing material.

Record specialist review and the purchasing rationale.

A missing documentary match is a review prompt. It does not prove that a supplier lacks the safeguard.

Research question and contribution

The semantic model is the contribution. The checker is the demonstration.

RESEARCH QUESTION

To what extent can semantic web technologies support ethical and legal compliance assessment in the procurement of LLM-based forensic tools?

Scope of this talk

Article 15 evidence requirements, semantic traceability and a small procurement demonstration.

1. Traceability model

Legal anchors linked to competency questions, safeguards, evidence indicators and review outcomes.

2. Operationalisation

Represent selected legal requirements as inspectable procurement evidence expectations.

3. Inspectable artefacts

RDF/OWL, SHACL, SPARQL, ontology documentation and the checker report.

Boundary: no legal certification, no proof of forensic accuracy and no replacement for expert judgement.

Legal anchors and applicability

The model organises review requirements. It does not automatically decide which law binds a procurement.

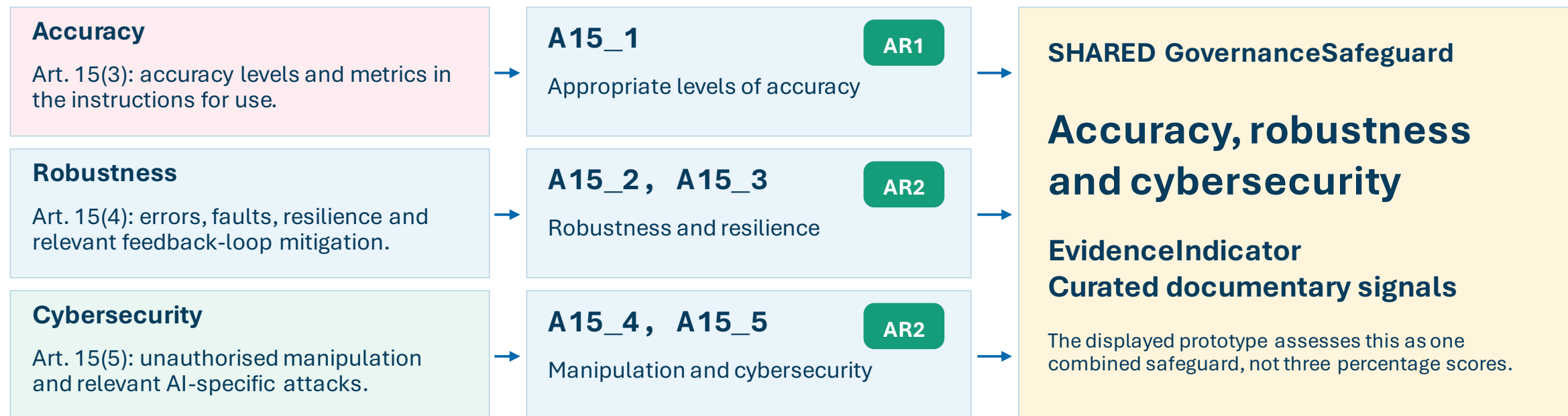
Anchor	Scope	Role in the framework
EU AI Act: Article 15	Accuracy, robustness and cybersecurity for high-risk AI systems.	Detailed mapping on the next slide.
EU AI Act: wider context	Articles 9–14, Article 27 and Annexes III/IV.	Risk management, data, documentation, logging, oversight and relevant rights-assessment context.
Public procurement law	Directive 2014/24/EU: Articles 18(1), 67 and 84.	Principles, award criteria and documentation of procurement decisions.
Data protection and rights	GDPR and the EU Charter of Fundamental Rights.	Privacy, fairness, non-discrimination and remedy as relevant review concerns.

Check jurisdiction, intended use and operator role first. “Forensic tool” or “uses GPAI” does not itself establish AI Act high-risk status.

Article 15 mapped into MTALO

Official legal paragraphs and internal model identifiers are different numbering systems.

Article 15(1): appropriate accuracy, robustness and cybersecurity, with consistent lifecycle performance.



AR1 asks whether accuracy levels and metrics are specified. AR2 asks about robustness, safety and cybersecurity measures.

Art. 15(2) supports benchmark development for accuracy and robustness. A15_1...A15_5 are saved MTALO identifiers, not statutory paragraph numbers.

TAF, MTALO and the checker

Three distinct roles. One traceable procurement review.

TAF

Trustworthy AI Forensics

The governance framework.

Structures review of procurement evidence, safeguards and human judgement.

MTALO

Mapping Trustworthy AI to Law Ontology

The semantic model.

Represents legal anchors, questions, safeguards, evidence and risk concepts.

Checker

Prototype demonstration

The implementation.

Uses deterministic screening to expose documentary matches, gaps and traceable outputs.

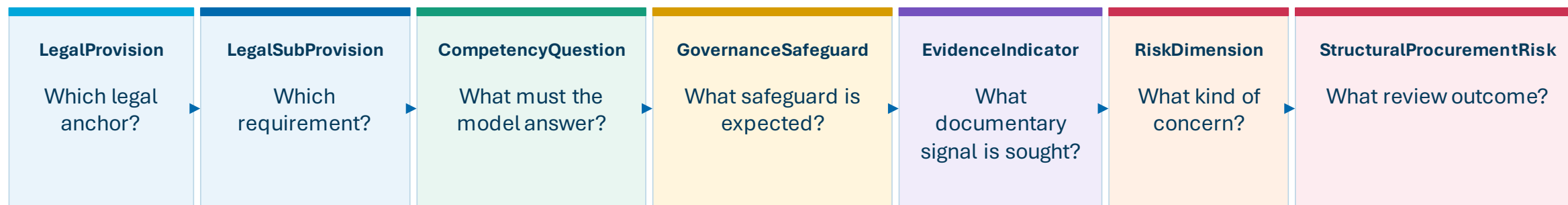
ONTOLOGY DEVELOPMENT: Linked Open Terms (LOT)



LOT guides ontology engineering. It is not the checker's legal applicability test or risk-scoring rule.

Ontology contribution: the complete traceability chain

The class names are preserved. The captions explain their role for a procurement reviewer.



EvidenceIndicator

A documentary signal associated with a safeguard. A detected term is not proof that the control works.

RiskDimension

A category of concern. It explains why a gap may matter, such as a legal obligation or an operational process. It is not the overall rating.

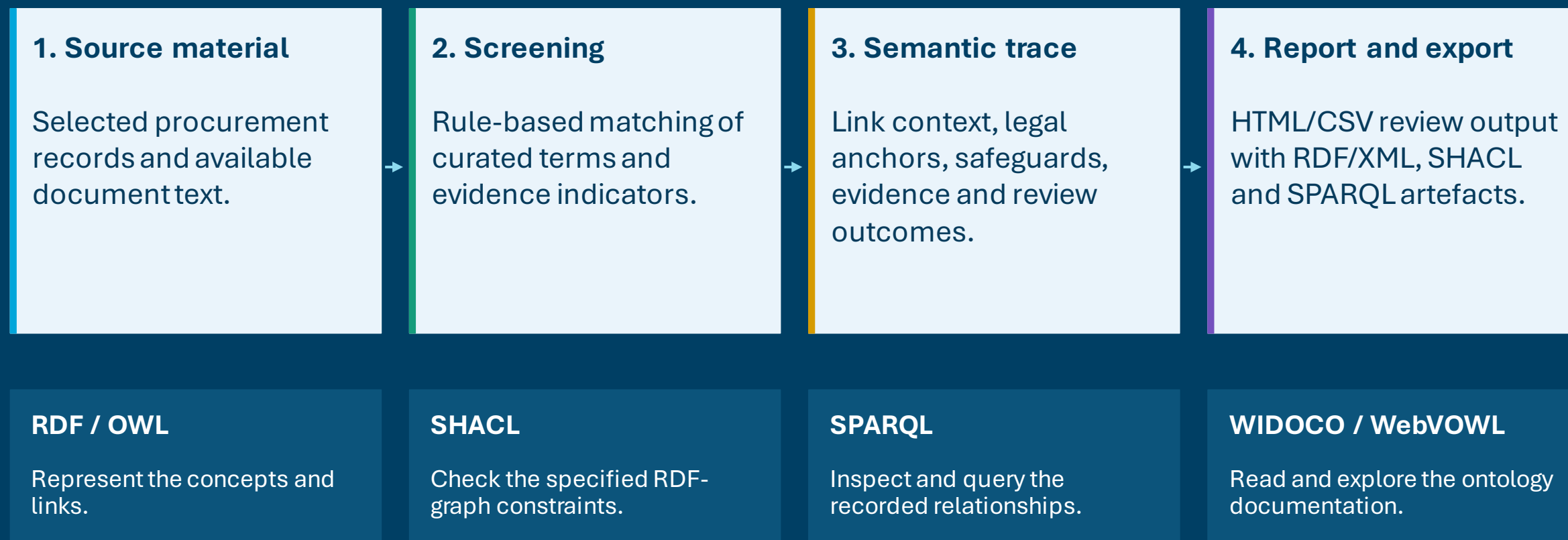
StructuralProcurementRisk

The contract-level review outcome under the checker rules. This is not the AI Act's "high-risk" classification.

Not evidenced does not mean absent in reality. A risk label does not establish legal or forensic validity.

TAF artefact pipeline: data to reviewable outputs

Separate record discovery, documentary matching and semantic representation.



A text-screening category is not a procurement use case or a verified product capability.

Richer NLP-assisted extraction is under development. No dataset totals are combined across different saved runs.

Prototype results by procurement context

Illustrative results from a saved checker report. Coverage is documentary alignment, not legal conformity.

Scenario assigned in report	Coverage	Safeguards evidenced	Missing / review gap	Reported risk
UC1: Provider → Deployer using GPAI	100%	8/8	None recorded	MODERATE
UC2: Provider → Deployer without GPAI	87.5%	7/8	Forensic Chain of Custody & Data Integrity	HIGH
UC3: Deployer procures GPAI directly	100%	8/8	None recorded	MODERATE

What this demonstrates

High documentary coverage can coexist with a HIGH review outcome. UC2 has one unevidenced core safeguard in the displayed snapshot.

What reviewers do next

Inspect the source passage. Request clarification. Record the specialist judgement.

Thank you. Q&A

TAF structures review. MTALO provides meaning. People make the procurement decision.

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Live Demo

Website: tranners.com/mcc_ai_checker_output.html

Ontology: tranners.com/Widoco/TAF_Ontology_Documentation.html

URI: w3id.org/taf

LinkedIn

